

telecommunications, broadcasting and navigation capabilities like never before even at places that you might not have heard of! The project essentially involves putting up an airship midway between the flying heights of drones and the satellites to do the job. Having a lifespan of approximately five years, these airships will offer better endurance and will be stationary too. Payloads of about 200kg can be carried by them.

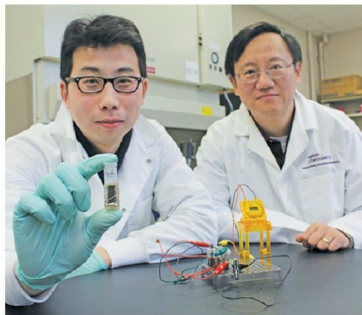
The ships will be positioned at an altitude of about 20km in the stratosphere and will be able to capture solar power in all directions at all times. Solar energy will charge an ultra-light fuel cell for energy storage and to power all the equipment on board. The developers are expected to roll out the first prototype in the next five years.

Sugar-powered batteries

Researchers at Virginia Polytechnic Institute and State University have developed a new bio-battery that is cheap, biodegradable and, quite interestingly, runs on sugar! Using sugar as fuel, the battery is powerful enough to run electronic gadgets and can run your smartphone for ten days, claim the developers.

The bio-battery works on a principle that closely resembles metabolism of food inside the human body, during which sugar is broken down into carbon dioxide and water, releasing electrons in the process. What sets it apart from other bio-batteries is the fact that it completely converts sugar into energy thereby generating greater power output.

In terms of functioning, the bio-battery utilises two active enzymes that release two pairs of electrons from the sugar. At the same time, ten other enzymes help to reset the reaction inside the battery. The active enzymes release another quartet of electrons following a reset. In this way, the battery is able to extract all the energy from a single molecule of sugar at the end of six cycles.



Sugar-powered battery (Courtesy: Virginia Tech University)

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Midget Flange cap



Midget Groove cap



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Wedge base

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- High reliability of indication
- One-time Expense



Binay Opto Electronics Pvt. Ltd.

44, Armenian Street, Kolkata 700 001, India
Telephone : (033) 2210 2039, 2210 3907, 4006 9875
email : info@binayled.com
Fax : 91-33- 2242 1493 <http://www.binayled.com>

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